Healthcare Provider Fraud Detection Project Report

This report summarizes a healthcare claims fraud detection project using provider-level claims, beneficiary, billing, demographic, chronic-condition, and risk-based features.

1. Project Objective

The objective of this project is to identify healthcare providers with elevated potential fraud risk using historical claims and beneficiary-level data. The model is designed as a fraud-risk screening tool, not as proof of fraud.

2. Test Prediction Summary

Metric	Value
Total test providers	1353
Predicted potential fraud providers	227
Predicted not potential fraud providers	1126
High risk providers	16
Medium risk providers	168
Low risk providers	1169

3. Data Used

The project used provider-level, claims-level, and beneficiary-level healthcare datasets. The provider dataset contained billing and utilization summaries. The claims dataset contained individual inpatient and outpatient claims. The beneficiary dataset contained patient demographic and chronic-condition information.

4. Data Cleaning

The claims data was cleaned by converting claim start and end dates into datetime format, checking invalid claim durations, handling missing deductible amounts, and validating negative reimbursement or deductible values. Beneficiary data was cleaned by converting date of birth and date of death fields into datetime format, encoding chronic-condition indicators, and encoding renal disease indicators. Provider-level data was checked for duplicates, missing values, and column-name consistency.

5. Feature Engineering

Provider-level features were created from claims and beneficiary data. These included demographic features such as average patient age, gender distribution, and race distribution; chronic-condition features such as average chronic disease burden and disease-specific rates; and fraud-risk features such as claim duration, high-cost claim percentage, reimbursement intensity, and inpatient claim ratio.

6. Rule-Based Risk Levels

A rule-based risk score was created using selected high-risk indicators. Providers received one point for exceeding training-data thresholds on features such as total claims, total reimbursement, claim-per-patient ratio, reimbursement

per patient, inpatient claim ratio, high-cost claim percentage, and maximum claim duration. Providers were categorized as Low, Medium, or High risk based on their total risk score.

7. Model Training and Comparison

Four machine learning models were trained and compared: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. The models were evaluated using fraud precision, fraud recall, fraud F1-score, ROC-AUC, PR-AUC. Since fraud detection is an imbalanced classification problem, recall, F1-score, and PR-AUC were especially important.

Model	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression	0.47	0.89	0.62	0.9669	0.7892
Decision Tree	0.43	0.87	0.58	0.8752	0.6626
Random Forest	0.52	0.88	0.65	0.9658	0.7635
Gradient Boosting	0.51	0.9	0.65	0.9674	0.7825

8. Selected Model

Select Model: Gradient Boosting

Reason:
Gradient Boosting was selected because it achieved the strongest overall fraud-screening performance on validation data.

Validation Performance:
- Fraud precision: 0.51
- Fraud recall: 0.90
- Fraud F1-score: 0.65
- ROC-AUC: 0.9674
- PR-AUC: 0.7825

Interpretation:
The model caught 91 out of 101 fraud-labeled providers in the validation dataset and missed only 10 fraud-labeled providers. It also had a strong ROC-AUC and PR-AUC, making it a strong model for provider fraud-risk screening.

Important Note:
This model should be used as a risk-screening tool, not as proof of fraud.

9. Test Prediction Results

The selected Gradient Boosting model was applied to 1353 actual test providers. The model predicted 227 providers as potential fruad and 1126 providers as not potential fraud.

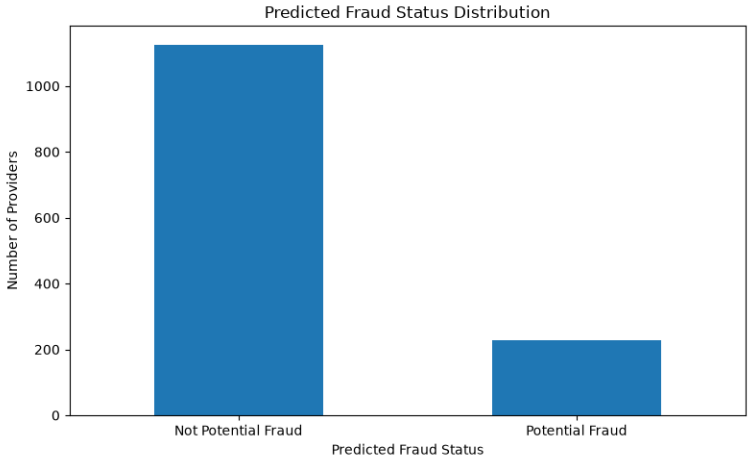
Risk Level	Not Potential Fraud	Potential Fraud
High	0	16
Medium	42	126

Low	1084	85

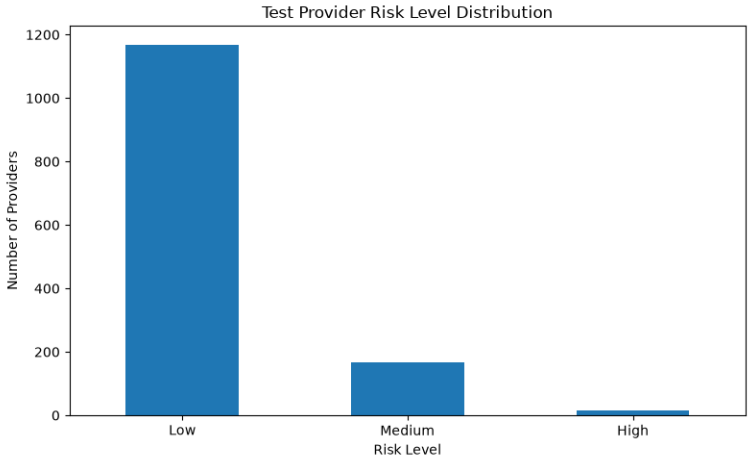
10. Project Visualizations

The following charts summarize the final test predictions, risk levels, model behavior, and model comparison results.

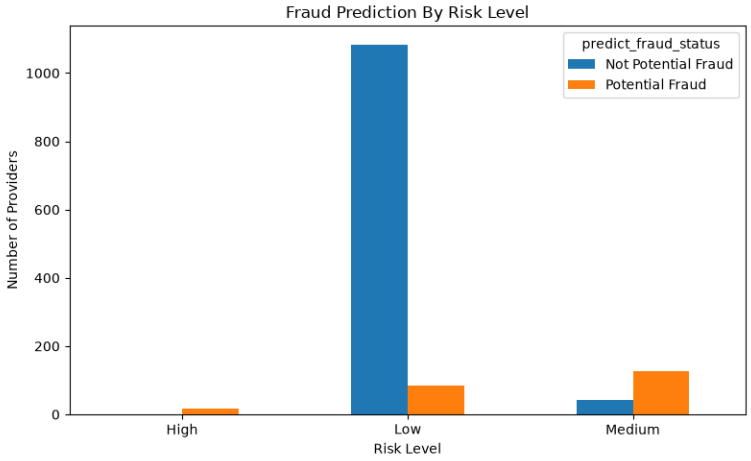
Predicted Fraud Status Distribution



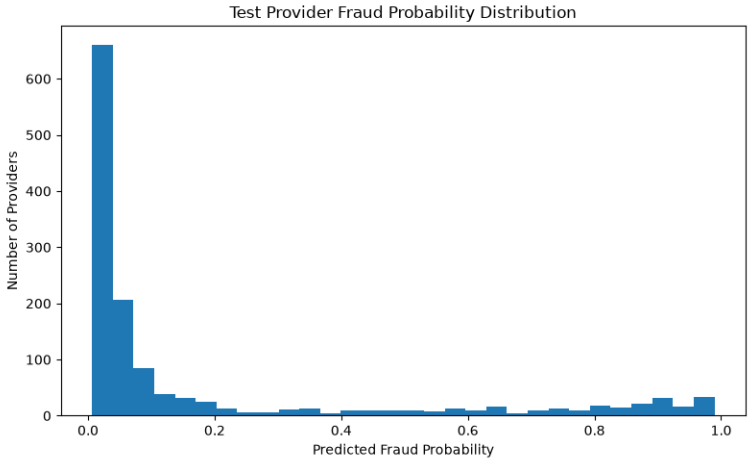
Risk Level Distribution



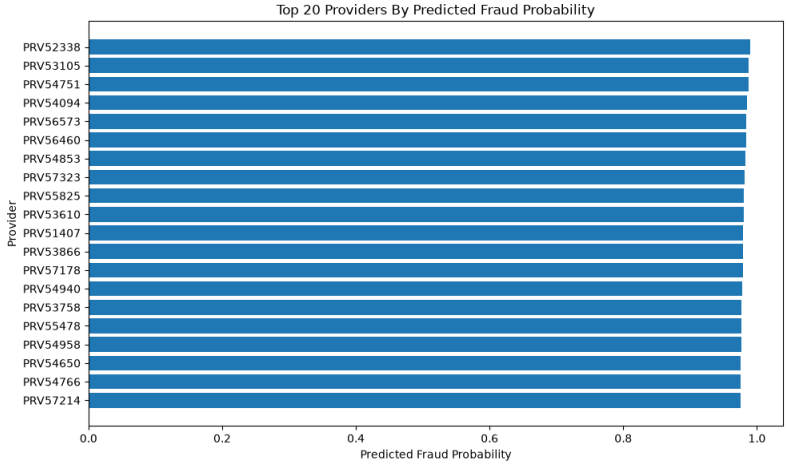
Fraud Prediction By Risk Level



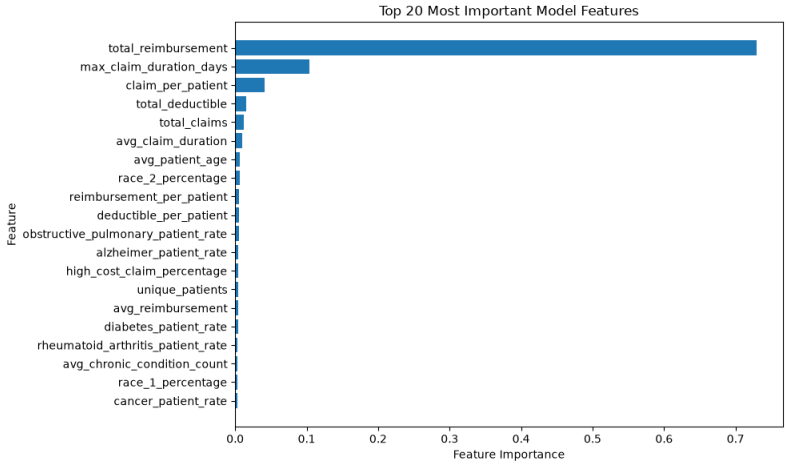
Fraud Probability Distribution



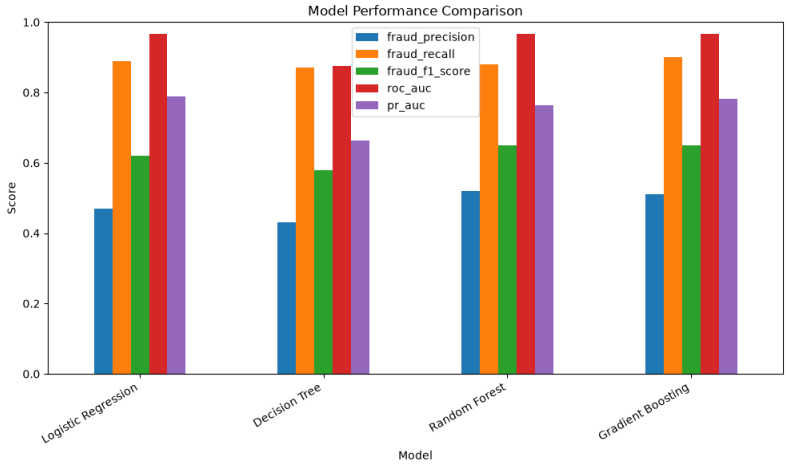
Top 20 Providers By Fraud Probability



Top 20 Model Feature Importance



Model Performance Comparison



11. Limitations

This project identifies potential fraud risk, not confirmed fraud. The dataset does not contain investigation outcomes or legal findings. The model is based on available claims, reimbursement, provider, demographic, and chronic-condition features. Diagnosis and procedure-code level fraud patterns could not be analyzed because those fields were not available in the combined claims dataset used for this project. Model predictions should therefore be used as a screening tool for prioirtizing review, not as final evidence of fraud.

12. Conclusion

The final Gradient Boosting model was seleted because it provided strong validation performance, especially fraud recall and fraud F1-score. On the actual test provider dataset, the model predicted 227 out of 1353 providers as potential fraud. The rule-based risk-level analysis also helped explain the model output by showing how predicted fraud providers were distributed across Low, Medium, and High risk categories.

Overall, the project demonstrates an end-to-end healthcare fraud detection workflow covering data cleaning, feature engineering, exploratory analysis, risk scoring, machine learning model training, test-data prediction, visualization, and reporting.